Additional file 1

How accurate is altitude from pressure geolocators? Supplementary methods and results

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This supplement provides reference-data rules and full-height error summaries (S1), environmental profiles (S2), the ERA5 product comparison (S3), reference-height and weighting checks (S4), and a horizontal-position pilot (S5). It uses the main manuscript's three conversions and error definitions: error is estimated minus reference height; station bias is mean signed error; SD is within-station variability; and p95 is the 95th percentile of individual absolute errors. **ERA5 standard** denotes the ERA5 surface reference with the standard lapse rate; **ERA5 virtual + GPT3** adds virtual temperature and a climatological lapse rate. Code, input provenance and extended diagnostics accompany the analysis [1].

S1 Reference data, selection and error distributions

Reference data and matching

GHCNh supplies ground pressure and reported station elevation for each station's latest qualifying year in 2022–2026 [2]. HadISD locations define the initial candidate footprint only. One randomly selected station per 2° cell is supplemented by stations above 1000 m; after screening, each retained cell receives equal weight, divided among its stations. GNSS barometers provide a complementary ground check in 2022–2024 [3, 4], and IGRA2 soundings provide the flight-height reference in 2023–2024 [5–7]. MeteoSwiss methods and results are in S4. Missing reference observations are not filled.

All observations are matched to ERA5 single-level surface pressure, 2 m temperature, dewpoint and orography at the nearest 0.25° grid cell and hour [8, 9]. GHCNh reports use the nearest hour (half-hour ties go to the earlier hour), retaining the first report per hour. GNSS samples within ±1 min of an hour are averaged to unique station-hours. Radiosonde level times combine release and elapsed ascent times; missing values use nominal release time or a 5 m s^{-1} ascent estimate. All levels use launch-station coordinates, ignoring balloon drift.

IGRA2 reports geopotential heights; these do not establish independent geometric-height accuracy at every level. We retain levels 1–6000 m above each sounding's surface, substituting station elevation if surface height is missing. Source extraction requires pressure $\geq$ 400 hPa, so vertical coverage also depends on launch elevation. Table 4 gives support by height band; no additional station filter is applied within bands.

Selection and quality control

Table 1 gives the retained rules. Listed station elevation may differ from barometer height [10, 11], and the ERA5-residual-based screen cannot distinguish reference errors from model errors. Screened ground results are therefore conditional on selection; S4 tests its effect. The step check uses monthly medians from the selected year and up to two preceding years, with a month-of-year effect and one breakpoint. It requires at least 18 monthly medians spanning all 12 calendar months: 2023 of 2144 post-match coverage-qualified stations were testable. Other stations are untested, rather than demonstrated stable.

GNSS heights combine site-log marker/antenna heights and barometer offsets; EGM96 geoid undulation is subtracted from ellipsoidal heights. The median signed difference from the file-header height must be within ±1 m. This checks metadata consistency, without establishing metre-level absolute accuracy or independence from ERA5. Retained ground coverage is summarized in Table 2; partial 2026 records must pass the same reporting threshold as completed years (S4).

Table 1: Selection and quality rules. MAD is R's scaled median absolute deviation. Offset and observation flags use ERA5-standard discrepancies; they are not independent diagnoses of sensor faults.

Reference/check	Selection or quality rule
GHCNh coverage	Use each station's most recent qualifying year from 2022–2026 with ≥ 183 reporting days and ≥ 4 distinct accepted station-hours per reporting day on average
GHCNh pressure QC	Pressure in 300–1100 hPa; exclude suspect/error codes and lower-case QC flags; retain missing QC and accepted legacy flags
Ground observation flag	Discrepancy from the station median exceeds both 100 m and 10 scaled MAD
GHCNh station flags	Absolute mean offset >30 m + 10% of the absolute station–ERA5 elevation gap; fitted step magnitude ≥ 10 m with absolute t ≥ 10; or >1% gross observations
GNSS coverage	≥ 180 listed file-days across the study window, ≥ 2160 usable unique hours and the header/site-log height check
IGRA coverage	Candidate record spans the study years with a one-year margin; ≥ 100 soundings before thinning and ≥ 100 available soundings in post-processing, across the two years
IGRA observation flag	Surface-referenced discrepancy from the height-band median exceeds both 150 m and 10 scaled MAD; finite atmospheric inputs and a retained sounding surface are required

Table 2: Retained ground inputs: GHCNh selected years in 2022–2026 and complementary GNSS records in 2022–2024. Hour availability is the median percentage of all calendar hours across station-years, including empty station-years; it reflects reporting schedules as well as gaps. Every retained row is a unique station-hour.

Reference	Stations	Unique hours	Median availability (%)
GHCNh	1983	8,309,533	52.7
GNSS	49	1,124,815	99.4

Individual errors across the three conversions

Table 3 and Table 4 extend the main Results with individual-error distributions. Ground observation weights divide each station's spatial weight among its retained hours. Flight weights divide each station's weight among all retained levels, then normalize within each height band. All conversions use identical observations and weights. Flight errors subtract the same sounding's surface error and exclude absolute ground-reference offsets.

Table 3: Screened GHCNh individual ground errors: median and 95th percentile of absolute error (m). Complementary GNSS and MeteoSwiss results are in S4.

Conversion	Median |error| (m)	p95 (m)
Standard atm.	41.0	143.1
ERA5 standard	3.9	21.6
ERA5 virtual + GPT3	4.0	21.6

Table 4: Surface-referenced flight-height errors across the full evaluated range. Conversion columns give median absolute error (p95), in metres. Station and sounding counts are shared by all conversions; a sounding may contribute to multiple bands. The source pressure cutoff can reduce upper-height coverage at elevated launch sites.

Height band	Stations	Soundings	Standard atm.	ERA5 standard	Virtual + GPT3
1–100 m	600	323,497	3.8 (19)	3.0 (19)	2.9 (19)
100–250 m	584	285,269	6.1 (28)	3.0 (22)	2.9 (22)
250–500 m	530	226,896	16.7 (52)	4.7 (23)	3.9 (21)
0.5–1 km	628	632,600	30.5 (77)	7.8 (32)	5.6 (28)
1–1.5 km	623	555,894	54.9 (128)	14.4 (52)	10.0 (40)
1.5–2 km	578	283,060	69.8 (137)	18.3 (63)	12.0 (48)
2–3 km	565	496,949	98.1 (238)	33.7 (120)	25.6 (86)
3–4 km	597	396,761	148.0 (231)	44.9 (127)	33.9 (101)
4–5 km	507	246,251	194.6 (416)	71.0 (228)	64.6 (190)
5–6 km	642	666,775	242.4 (392)	94.4 (253)	80.1 (240)

S2 How the error varies

Ground: elevation gap, climate and time

The station–ERA5 elevation gap is associated with within-station scatter (Figure 1): fitted SD rises from 3.2 m at zero gap to 11.2 m at a 1000 m gap. This gap is distinct from terrain roughness, the sub-grid SD of orography used to classify terrain in the main manuscript. Seasonal, local-time and climate profiles provide context for the pooled ground summaries (Figure 2; Figure 3); the polar group has greater scatter.

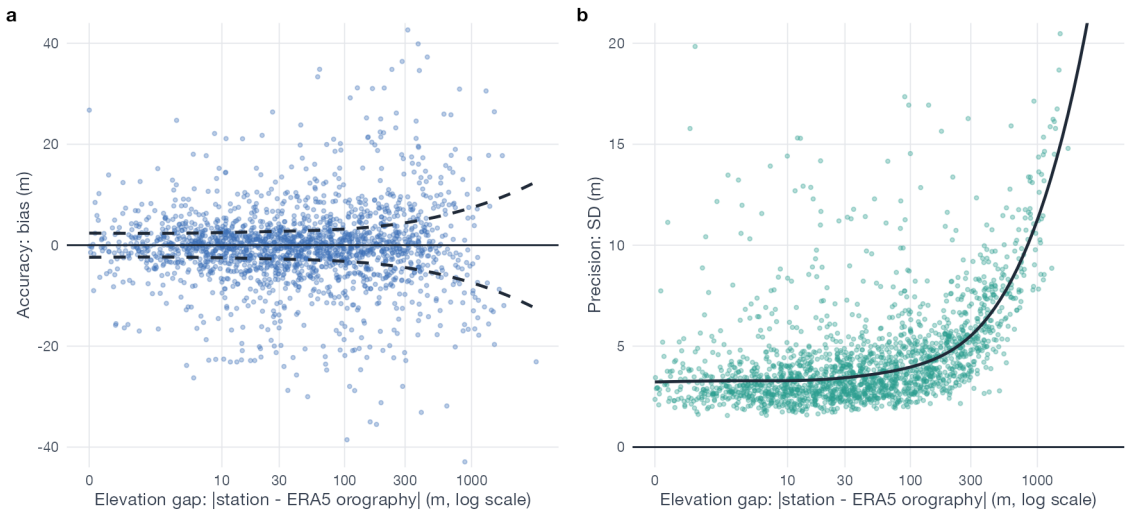


Figure 1: Bias (a) and SD (b) per station against the elevation gap between station and ERA5 orography (GHCNh reference set, ERA5 standard conversion). Dashed: ± fitted |bias|; solid: fitted SD. Cell-weighted GAMs use log(|bias| + 0.5 m) or log(SD) as the response and a smooth of log(1 + absolute elevation gap); curves are back-transformed fits. Gap and error values inside logarithms are expressed numerically in metres.

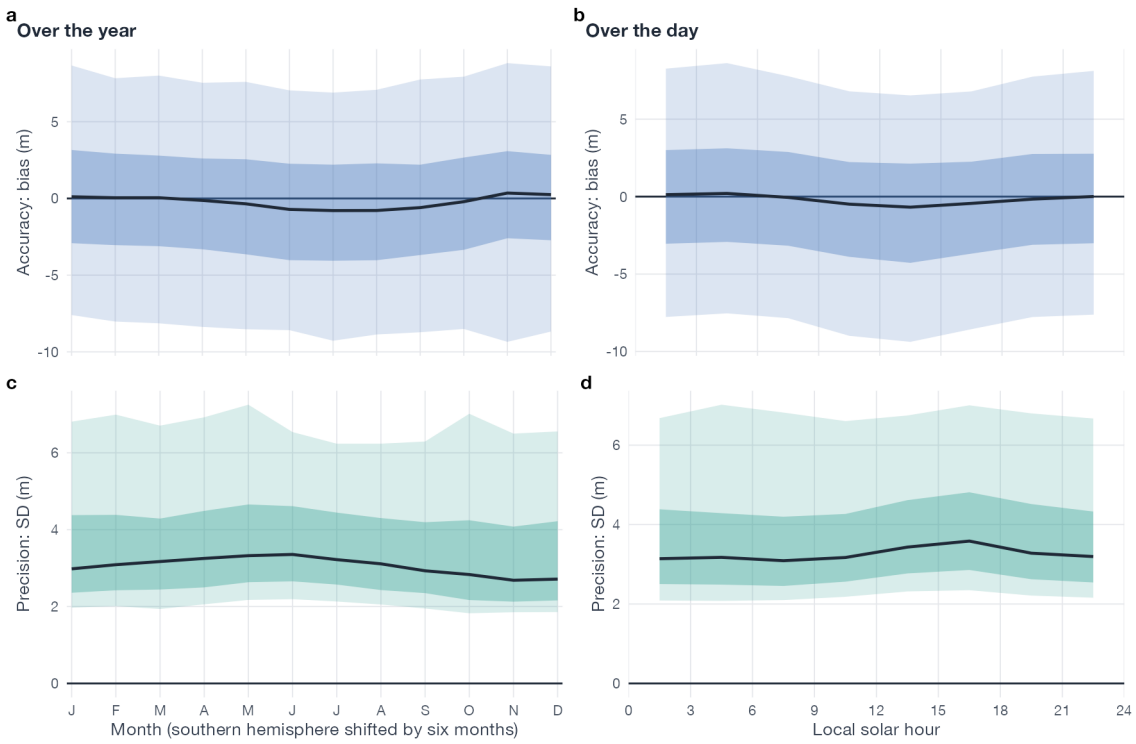


Figure 2: Station bias (a, b) and SD (c, d) by seasonal month (southern hemisphere shifted by six months) and 3 h local-solar-time band, screened GHCNh, ERA5 standard conversion. Station–time classes require at least 30 observations. Lines: cell-weighted station medians; bands: station 25–75% and 10–90% ranges, rather than confidence intervals.

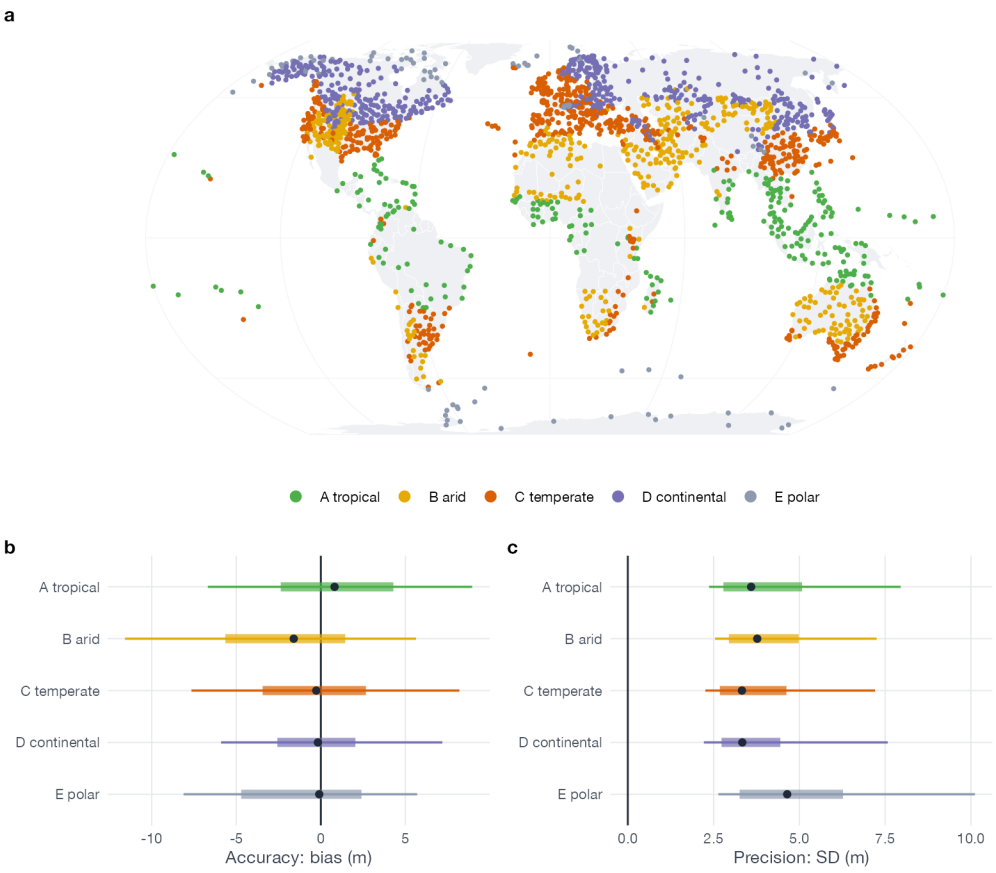


Figure 3: Ground errors by Köppen-Geiger climate zone, screened GHCNh, ERA5 standard conversion. (a) Station locations; (b) signed station bias; (c) within-station SD. Points: cell-weighted station medians; thick and thin segments: 25–75% and 10–90% station ranges.

Only 5 stations contribute to the paired historical comparison (Figure 4). Their discrepancies decline across the sampled years, but this small common-station subset cannot establish a global accuracy trend.

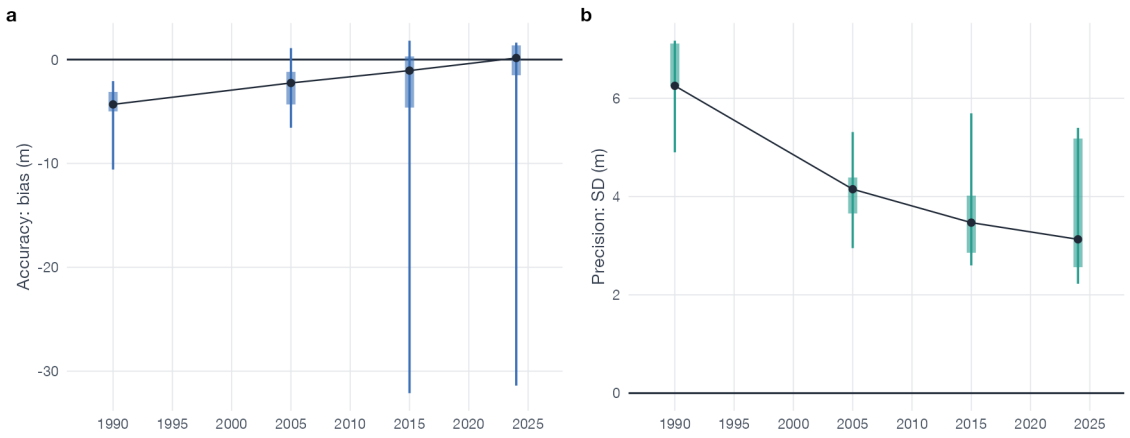


Figure 4: Station bias (a) and SD (b), ERA5 standard conversion, at the same five GHCNh stations in 1990, 2005, 2015 and 2024 (at least 500 usable observations per station-year). Points: station medians; thick and thin segments: 25–75% and 10–90% station ranges; lines connect medians. The sample is descriptive and geographically restricted.

Flight: climate, season and column temperature

Climate and half-year profiles show why a pooled signed error can conceal opposing biases (Figure 5). The corrected conversion reduces these seasonal biases, while a climatological lapse rate cannot reproduce every sounding's profile. For ERA5 standard, the error predicted from the assumed versus observed mean column temperature closely follows the surface-referenced error ($r = 0.97$; Figure 6). Night-time underestimation is generally greater than daytime underestimation (Figure 7), consistent with surface inversions making the column warmer than the surface-based profile assumes.

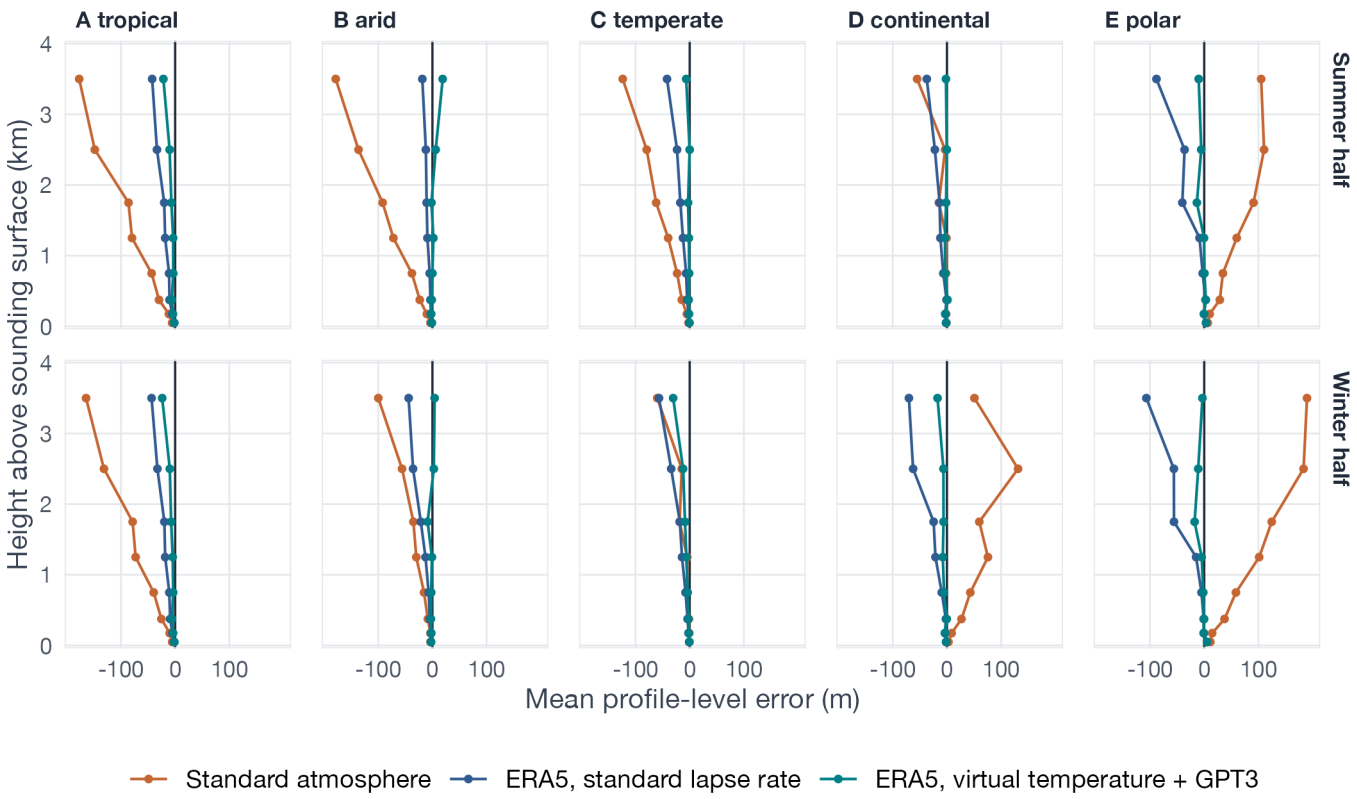


Figure 5: Mean signed surface-referenced profile-level error by conversion, Köppen-Geiger climate zone and half-year, shown to 4 km. These are pooled level summaries using the primary flight weights, rather than medians of station biases. Summer half is 21 March–22 September in the northern hemisphere and shifted by six months in the southern hemisphere; classes with fewer than three stations are omitted.

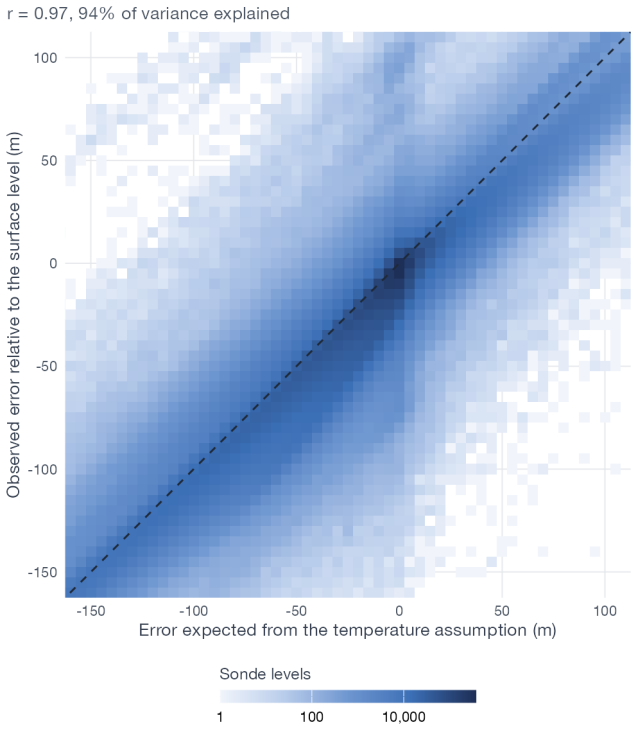


Figure 6: Observed surface-referenced flight-height error against the error predicted from the assumed versus observed mean column temperature, ERA5 standard conversion, at radiosonde levels more than 200 m above the sounding surface. Tile colour: number of levels (log scale); dashed line: 1:1 agreement. The displayed window is −150 to 100 m on both axes; the reported correlation uses the full retained comparison. This is a mechanism check using the same atmospheric profiles, rather than an independent validation.

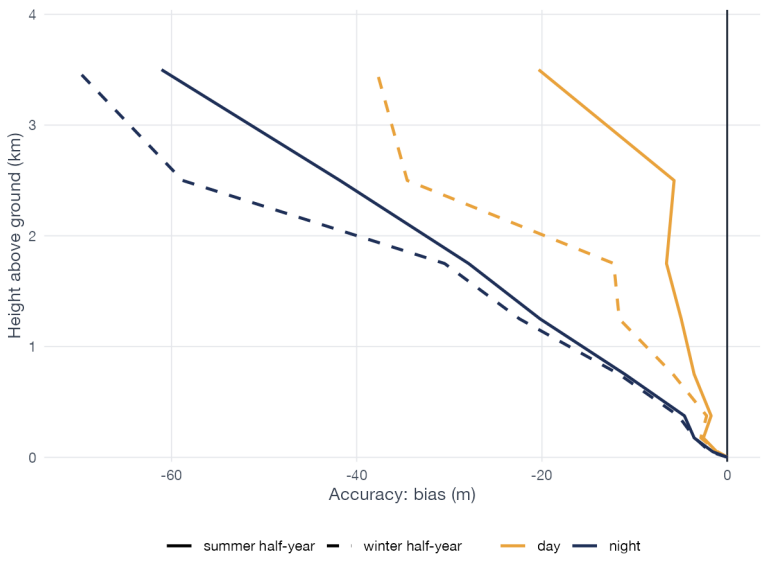


Figure 7: Mean signed surface-referenced profile-level error for ERA5 standard by day/night and half-year, shown to 4 km with the primary flight weights. Day is 07:00–19:00 local solar time; all other hours are night. Negative error indicates underestimated height. Half-years use the same hemispherically adjusted dates as the climate profiles; curves join observed height-band summaries.

S3 ERA5 single-levels and ERA5-Land

ERA5-Land's finer 0.1° grid [12] does not reduce persistent ground offsets in these comparisons. Median station |bias| is 9.9 m versus 2.9 m for ERA5 single-levels at GHCNh stations, with the same ordering at GNSS and MeteoSwiss stations (Figure 8). Surface-referenced flight errors are similar (Figure 9). Together, these results point to a difference in the surface reference; they do not by themselves identify an inconsistency in ERA5-Land's pressure adjustment or published orography.

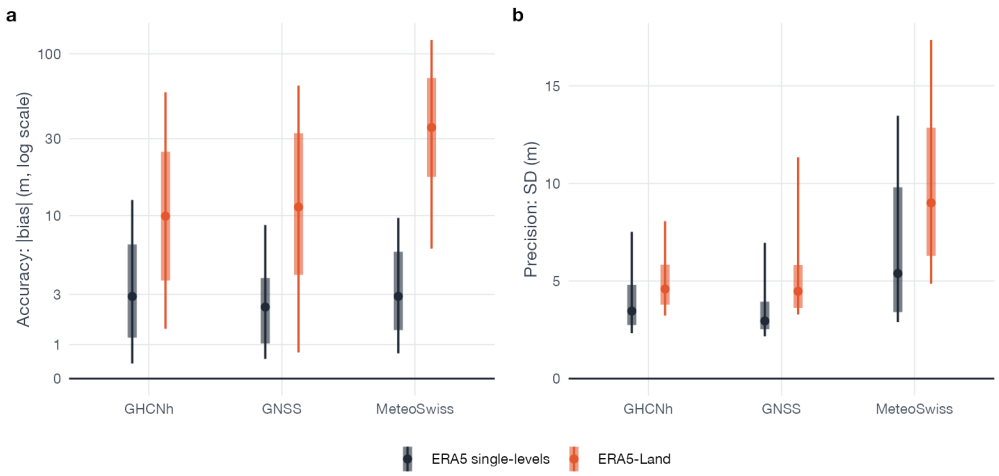


Figure 8: Station |bias| (a, pseudo-log scale) and SD (b) with ERA5 single-levels and ERA5-Land, using the standard lapse rate at the same stations within each ground reference. Points: weighted station medians; thick and thin segments: 25–75% and 10–90% station ranges. GHCNh uses cell weights; GNSS and MeteoSwiss use equal station weights.

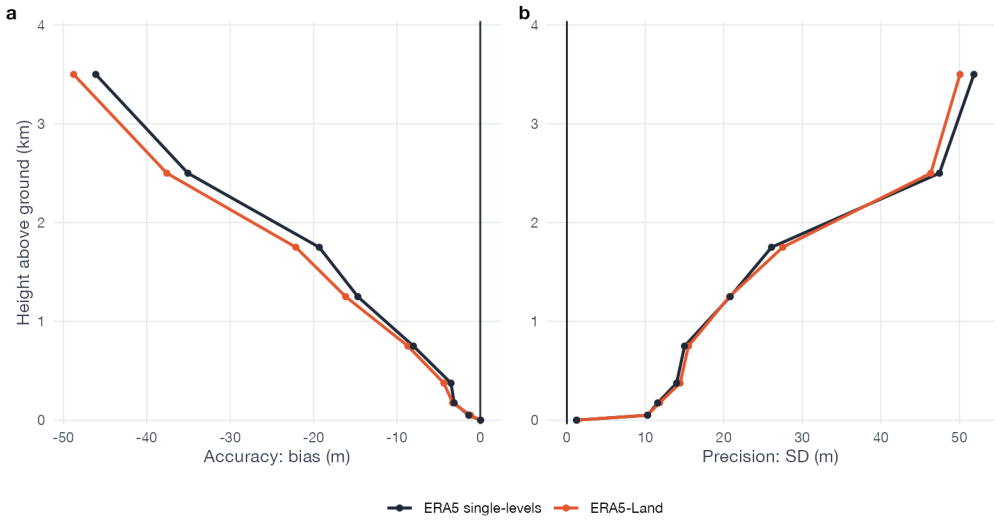


Figure 9: Pooled surface-referenced flight-height bias (a) and SD (b) by height band, ERA5 single-levels versus ERA5-Land with the standard lapse rate. Curves use the primary level weights and are shown to 4 km. Pooled SD includes between-station differences and differs from the median within-station SD in the main manuscript.

S4 Reference-height uncertainty and sensitivity

MeteoSwiss corroboration

SwissMetNet supplies 10-min instantaneous pressure at published barometer height (QFE, `prestas0`) [13]. We use finite full-UTC-hour values in 2022–2024 at 138 stations with a published barometer height and at least 2160 usable hours. Published sea-level heights are used without a Swiss-to-EGM96 transformation. Source: MeteoSwiss.

At 24 matched stations (Figure 10), replacing only the listed GHCNh reference elevation with the published barometer height reduces median station |bias| from 6.0 to 3.0 m, close to 2.7 m using MeteoSwiss pressure. Only the height-only contrast holds pressure and ERA5 series fixed: GHCNh uses its selected year, whereas MeteoSwiss covers 2022–2024. This supports a metadata contribution at these sites, without identifying the cause of discrepancies elsewhere. Table 5 summarizes the complementary ground references.

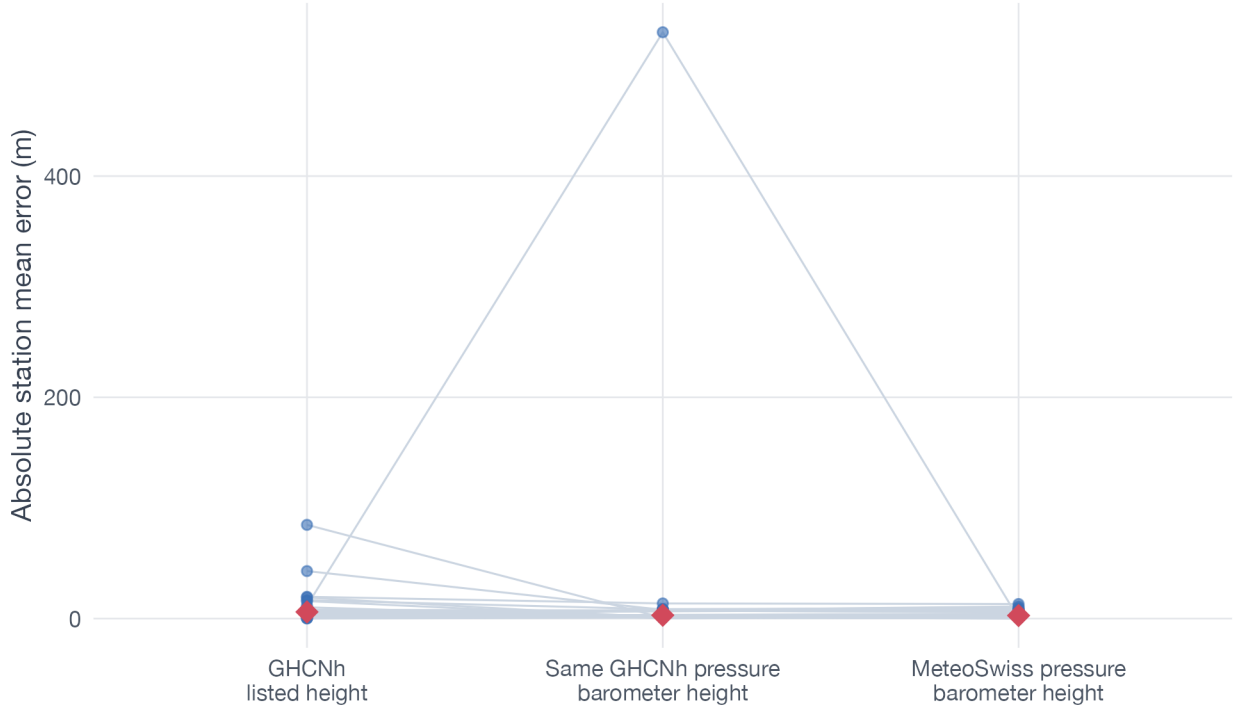


Figure 10: Absolute station biases with the ERA5 standard conversion at matched Swiss stations: listed GHCNh height, a height-only change using the same GHCNh pressure records, and MeteoSwiss pressure with published barometer height. Lines link stations; diamonds mark medians.

Table 5: Complementary ground-reference discrepancies (m), GNSS and MeteoSwiss. Station bias/SD describe persistent offset/scatter; median absolute error and p95 describe weighted individual errors.

Reference	Conversion	Median |bias|	Median SD	Median |error|	p95
GNSS	Standard atm.	22.6	74.0	50.3	162.0
GNSS	ERA5 standard	2.4	3.0	3.0	19.2
GNSS	ERA5 virtual + GPT3	2.4	3.0	3.0	18.9
MeteoSwiss	Standard atm.	41.1	68.0	60.2	159.5
MeteoSwiss	ERA5 standard	2.9	5.4	4.2	22.8
MeteoSwiss	ERA5 virtual + GPT3	2.6	5.8	4.4	22.6

Reference-plausibility screen

Removing the ERA5-residual-based station screen changes central errors less than their upper tails (Table 6). The step/noise-only alternative removes the offset criterion but keeps the step-change and frequent-gross-observation checks from S1. None of these flags identifies the cause of a residual. Screened values should therefore be read alongside the all-eligible comparison, rather than as unconditional conversion accuracy.

Table 6: GHCNh selection sensitivity, with cell weights assigned after each selection. Isolated gross observations and incomplete atmospheric inputs are excluded throughout.

Selection	Conversion	Stations	Median |error|	p95	Median |bias|
All eligible	Standard atm.	2144	42.5	164.9	23.8
All eligible	ERA5 standard	2144	4.3	40.1	3.2
All eligible	ERA5 virtual + GPT3	2144	4.3	39.9	3.2
Step/noise only	Standard atm.	2068	42.2	155.7	23.4
Step/noise only	ERA5 standard	2068	4.2	33.4	3.1
Step/noise only	ERA5 virtual + GPT3	2068	4.2	33.3	3.1
Screened	Standard atm.	1983	41.0	143.1	22.8
Screened	ERA5 standard	1983	3.9	21.6	2.9
Screened	ERA5 virtual + GPT3	1983	4.0	21.6	2.9

Flight weights and height-band coverage

Primary flight weights give each station equal total weight across all its retained levels; its contribution within a particular band therefore depends on vertical sampling. All 648 retained stations occupy distinct 2° cells, so equal cell and equal station weights are equivalent for this network. We compared equal station weights within bands, equal sounding weights within station–bands, and minimum coverage of 30 or 100 soundings per station–band. The corrected conversion retains the lowest flight-time-weighted median absolute error (Table 7), although height-specific upper tails vary (Table 8). Coverage thresholds change the contributing population; they do not establish more accurate references.

Table 7: Flight-time-weighted median absolute error (m), using unchanged biological height weights.

Weighting	Standard atm.	ERA5 standard	ERA5 virtual + GPT3
Primary	21.8	7.4	5.9
Equal station/band	19.8	6.9	5.4
Equal sounding/band	20.1	6.9	5.5
Station/band, minimum 30	20.0	6.9	5.4
Station/band, minimum 100	20.2	6.9	5.3

Table 8: Sensitivity of the 4–5 km absolute-error p95 (m) to within-band weighting and coverage. This band illustrates the upper-tail sensitivity concealed by an overall bird-weighted summary; biological height weights are not applied here. Minimum-coverage variants use equal station weights within bands.

Weighting	Standard atm.	ERA5 standard	ERA5 virtual + GPT3
Primary	416	228	190
Equal station/band	507	246	234
Equal sounding/band	538	265	259
Station/band, minimum 30	541	274	265
Station/band, minimum 100	528	285	245

Biological flight target and height weights

The biological target is the flight-time height distribution of 203 tags from 25 species (34,580 represented flight hours) in the Master GeoLocator Data Package, version 2026.0 [14]. Inputs match the published checksums. We retain most-likely pressurepath records with finite tag and surface pressure and no discard label; non-integer stationary-period identifiers mark flight. Upstream classifications and pressurepaths are used as supplied, without uniformly reprocessing the tracks.

Table 9 gives the height and duration assumptions needed to reproduce the weights. Main-manuscript Figure 4c shows the resulting height distribution. Alternative flight-duration, low-height and equal-tag targets preserve the conversion ranking (Table 10). Equal-tag weighting gives each contributing tag the same total weight, whereas the primary target pools represented flight time; neither is a global sample of birds. The two-hour threshold is based on represented sampling intervals, rather than directly observed take-off and landing.

Table 9: Construction of biological height weights. The pressure-offset adjustment and negative-height treatment do not establish true clearance above local terrain. All conversions are evaluated with the same biological weights.

Component	Primary rule
Represented duration	Interval to the next retained record of the tag, capped at 60 min; the last record uses the tag's median positive interval
Height conversion	Tag pressure relative to ERA5 surface pressure, with 288.15 K and the standard lapse rate
Stationary pressure offset	Interpolate between the median tag-minus-ERA5 pressure discrepancies in the preceding last six stationary hours and following first six stationary hours; use one endpoint alone, or zero if neither exists
Height limits	Residual negative estimates (8.5% of flight time) enter the lowest band; estimates above 6 km enter the highest evaluated band
Combining flight errors	Normalize radiosonde level weights within each height band, multiply by its share of bird flight time, and calculate the median of the combined absolute-error distribution

Table 10: Biological target sensitivity. Hours are represented flight time; median absolute errors are in metres. Equal-tag weighting retains the same records but normalises each tag's total weight to one. Standard atm.: fixed standard atmosphere; virtual: ERA5 with virtual temperature and GPT3.

Target	Tags	Hours	Standard atm.	ERA5 standard	Virtual
All flight	203	34,580	21.8	7.4	5.9
Flights ≥2 h	199	33,661	22.6	7.6	6.0
Points ≥50 m	203	30,058	28.3	8.8	6.7
Both restrictions	199	29,410	28.9	8.9	6.7
Equal tag	203	34,580	18.7	6.8	5.4

Removing the stationary pressure-offset adjustment, shifting all estimated heights by ±100 m, or changing the height formula on the subset with complete weather annotations also preserves the

ranking. These stress tests do not supply uncertainty intervals or make the weights independent of the conversion assumptions. Full results, annotation coverage, species contributions and sampling cadence accompany the code [1].

Reference dependence

ERA5 assimilates conventional surface and upper-air observations [9]; we did not establish whether each evaluated station, variable and date was assimilated. GNSS archive membership does not establish independence. The terrain-matched archive comparison (Table 11) and nearby-station counts (Table 12) are descriptive checks only: similar errors across archives, or at isolated stations, cannot quantify assimilation influence. Reference-height uncertainty and the residual-based screen remain additional limitations.

Table 11: Median |bias| and median SD (m) by terrain roughness (sub-grid SD of the ERA5 orography), ERA5 standard conversion.

Set	Terrain	Stations	median |bias|	median SD
GHCNh	flat (<20 m)	721	2.4	3.0
GHCNh	hilly (50-150 m)	483	3.5	4.0
GHCNh	gentle (20-50 m)	368	3.0	3.2
GHCNh	rough (150-300 m)	304	5.5	5.6
GHCNh	mountain (>300 m)	107	7.3	7.7
GNSS	hilly (50-150 m)	19	1.9	3.2
GNSS	gentle (20-50 m)	8	1.2	2.7
GNSS	mountain (>300 m)	4	6.9	8.0
GNSS	flat (<20 m)	15	2.5	2.5
GNSS	rough (150-300 m)	3	5.3	3.9
MeteoSwiss	mountain (>300 m)	69	5.1	9.7
MeteoSwiss	rough (150-300 m)	44	1.8	4.1
MeteoSwiss	hilly (50-150 m)	25	2.4	2.9

Table 12: GHCNh stations in flat terrain (sub-grid orography SD below 20 m), grouped by the number of other GHCNh stations within 300 km. Errors are in metres, ERA5 standard conversion; nearby-station counts do not identify assimilated observations.

Neighbours within 300 km	Stations	median |bias|	median SD
0-2	208	2.4	3.2
3-10	235	2.6	3.0
11-30	120	2.5	2.9
>30	158	1.6	3.0

Choice of ground observation year

The latest coverage-qualified year is selected by recency, without ranking years by ERA5 error. Selected-year station counts are 2022: 35; 2023: 40; 2024: 177; 2025: 404; 2026: 1327. Partial 2026

records have incomplete seasonal coverage. Table 13 compares the selected year with the latest qualifying completed year on the same screened stations. Each paired year must retain at least 183 reporting days and four distinct accepted hours per reporting day after ERA5 matching; the station screen is held fixed from the selected year.

Table 13: Date-range sensitivity for ERA5-standard ground discrepancies, on the same paired stations. Completed years satisfy the reporting threshold but need not contain observations in every month. The station screen is held fixed from the selected year.

Choice	Stations	Days	Months	Median |error|	Median |bias|	Median SD
Completed	1912	362	12	3.97	2.96	3.44
Latest	1912	280	10	3.97	2.92	3.50

Processing provenance, including the paired GHCNh–HadISD transition audit, source-specific QC composition and repeated-GNSS-hour check, is retained with the code [1]. The primary results use unique GNSS station-hours and source-aware pressure QC as specified in S1.

S5 Sensitivity to horizontal position

Design. We shift the ERA5/GPT3 query position, keeping pressure, time and radiosonde reference height fixed. The fixed-seed pilot samples ten IGRA2 stations across five climate zones and two terrain strata ($<$100 or $\geq$ 100 m sub-grid orography SD; six northern and four southern stations). Each supplies two eligible launches nearest mid-quarter per quarter of 2024, using the level nearest 2 km (within 1.7–2.3 km) and its surface record. Displacements are 0, 10, 25, 50, 100 and 200 km, with eight bearings at positive distances. Small shifts can select the same ERA5 cell; the standard atmosphere is location-independent.

Interpretation. Absolute altitude and surface-referenced flight height respond differently (Figure 11). The latter subtracts the surface error at the same displaced position and original surface-observation time. Small changes in pooled median error can conceal larger individual paired changes (Table 14). This fixed-height pilot does not establish a global location-error distribution or acceptable displacement radius, and excludes terrain-clearance uncertainty, balloon drift, moving anchors and rare weather extremes.

Table 14: Atmospheric-query displacement with ERA5 virtual + GPT3 near 2 km (all errors in metres). Median error change is displaced minus exact-location median absolute error. Paired median and p95 summarize absolute changes between matched displaced and exact-location errors. Stations, launches and bearings receive equal weights.

Outcome	Distance (km)	Median error change	Paired median	Paired p95
Absolute altitude	50	0.2	5.1	21.1
Absolute altitude	200	4.2	12.5	54.1
Flight height	50	-1.1	4.9	22.6
Flight height	200	0.6	11.5	48.0

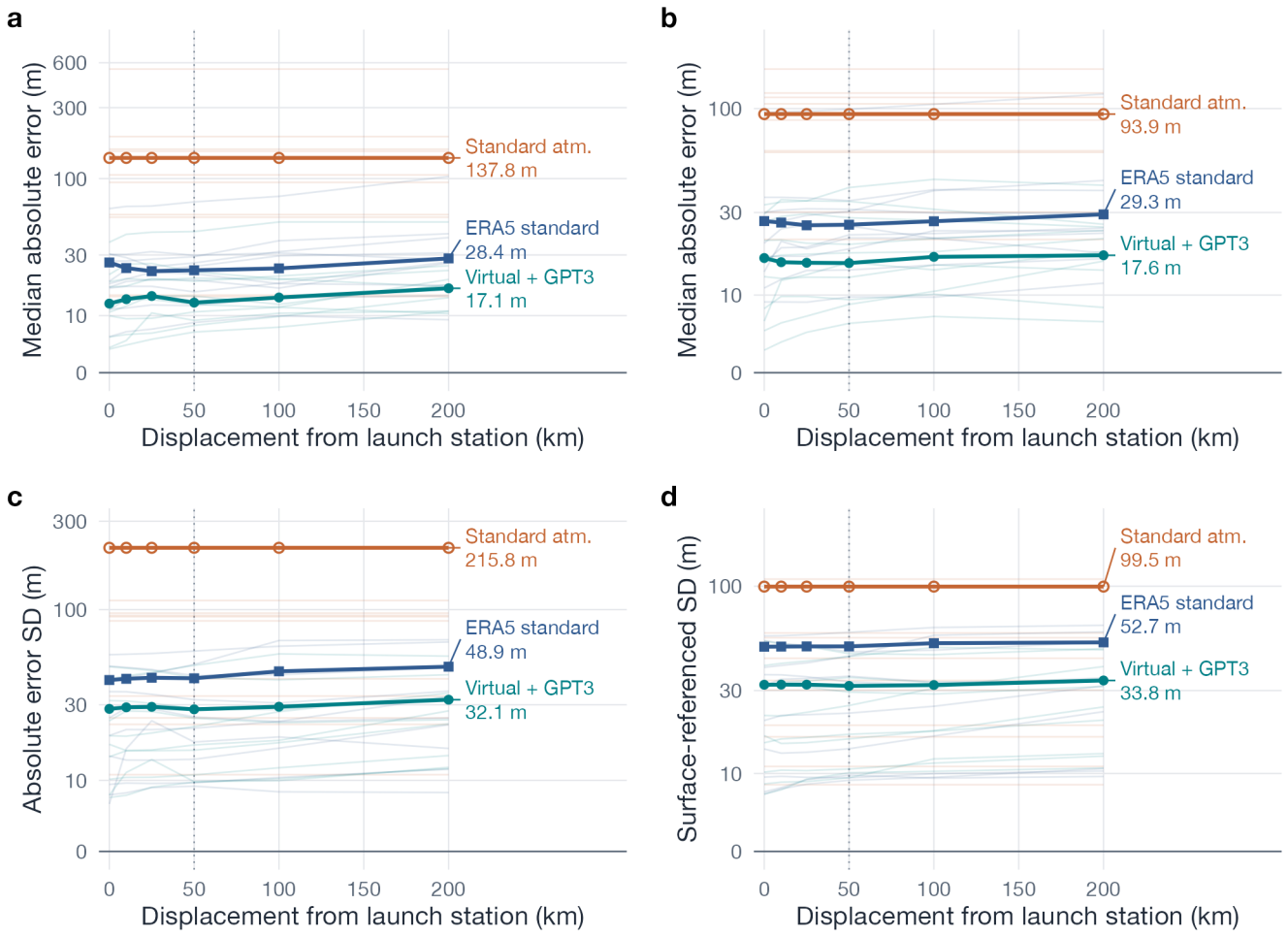


Figure 11: Error versus nominal displacement of the atmospheric query from the launch station. Left: absolute altitude near 2 km above the sounding surface; right: height after subtracting the displaced surface error. Upper panels: median absolute error; lower panels: pooled error SD. Faint curves: individual stations; strong curves and markers: pooled pilot (10 stations, 80 soundings, eight bearings per positive distance). Direct labels give pooled values at 200 km. The dotted vertical guide marks 50 km. Curves join evaluated distances without smoothing. All vertical axes use a pseudo-log scale (linear close to zero) so the station traces and pooled summaries remain visible together. Pooled SD includes between-station offsets; station scatter is empirical variation, not a confidence interval.

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